

Zheyang Wu, Ph.D.

Professor of Mathematical Sciences

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Research interests: Statistical methods for combining dependent evidence — p-value combination, signal detection, and integrative association testing — with applications to meta-analysis, whole-genome sequencing, statistical genetics, and clinical studies; statistical computing and open-source R software.

PROFESSIONAL EXPERIENCE

1. Education

- Ph.D. in Biostatistics, 2009, Yale University, New Haven, CT.
- M.Phil. in Epidemiology & Public Health, 2007, Yale University, New Haven, CT.
- M.S. in Mathematics, 2004, University of New Orleans, New Orleans, LA.
- B.S. in International Trade, 1998, Chongqing University, Chongqing, China.

2. Professorship at WPI

- 2021/09 – , Professor of Mathematical Sciences.
- 2015/09 – 2021/08, Associate Professor of Mathematical Sciences, Bioinformatics and Computational Biology, and Data Science.
- 2009/08 – 2015/08, Assistant Professor of Mathematical Sciences.

SCHOLARSHIP

3. List of publications or scholarly contributions

(Underlined authors are my advisees.)

Articles published on peer-reviewed journals:

- (1) David Clement, Emmanuel Agu, Atifa Sarwar, Deepak Kumar, **Zheyang Wu**, and Jean A. King (2026) "Machine learning Prediction of Self-Reported Real-World Knee Osteoarthritis Pain from Wearable Accelerometer-Derived Gait Features", *Scientific Reports*, accepted.
- (2) Shaoming Yin, Jianchang Lin, and **Zheyang Wu** (2026) "Augmenting clinical trial data with external controls through energy balancing weighted power prior", *Statistics in Pharmaceutical Research*, <https://doi.org/10.1080/19466315.2025.2587049>
- (3) Shaoming Yin, **Zheyang Wu**, and Jianchang Lin (2026) "Assumption-agnostic deep learning framework for holistic clinical trial monitoring", *Therapeutic Innovation & Regulatory Science*, Volume 60, pages 782-797 <https://doi.org/10.1007/s43441-026-00915-1>.

- (4) Xiaohui Chen, Fangfang Wang, Hong Zhang, and **Zheyang Wu** (2026) "On Correlation Robustness of Supremum-based p -Value Combination Tests", *Statistica Sinica*, accepted.
- (5) Hong Zhang, Ming Liu, John E. Landers, and **Zheyang Wu** (2026) "Integrated Weighted Association Test with Application to Genetic Association Studies", *Annals of Applied Statistics*, accepted.
- (6) Gonzalo Contador and **Zheyang Wu** (2026) "Optimal Adjustment and Combination of Independent Discrete p -Values", *Journal of Computational and Graphical Statistics*, accepted.
- (7) Shaoming Yin and **Zheyang Wu** (2026) "Evaluating cohort balancing methods for estimating marginal hazard ratios in small to moderate sample sizes", submitted to *Statistics in Medicine*.
- (8) Yanwen Zhu, Chen Zhao, Yishuo Xu, **Zheyang Wu**, Akiko Maehara, Liang Wang, Dirui Zhang, Ming Zeng, Rui Lv, Xiaoya Guo, Mengde Huang, Minglong Chen, Gary S. Mintz, Dalin Tang, Haibo Jia, and Bo Yu (2025) "Computational Modeling to Predict Conservative Treatment Outcome for Patients with Plaque Erosion: An OCT-Based Patient-Specific FSI Modeling Study", *Computer Modeling in Engineering & Sciences*. Published Online: 01 August 2025. DOI:10.32604/cmescs.2025.067039.
- (9) Gonzalo Contador and **Zheyang Wu** (2025) "A minimum Wasserstein distance approach to Fisher's combination of independent discrete p -values", *Scandinavian Journal of Statistics (SJS)*, 52(3):1281–1300 DOI: 10.1111/sjos.12787.
- (10) Yanwen Zhu, Chen Zhao, **Zheyang Wu**, Akiko Maehara, Dalin Tang, Liang Wang, Zhanqun Gao, Yishuo Xu, Rui Lv, Mengde Huang, Xiaoguo Zhang, Jian Zhu, Haibo Jia, Bo Yu, Minglong Chen, Gary S. Mintz (2024) "Comparison and Identification of Human Coronary Plaques with/without Erosion Using Patient-Specific Optical Coherence Tomography-Based Fluid-Structure Interaction Models: A Pilot Study", *Biomechanics and Modeling in Mechanobiology*, 1-19. DOI: 10.1007/s10237-024-01906-7.
- (11) Nan Lin, Shaoju Wu, **Zheyang Wu**, Songbai Ji (2024) "Efficient Generation of Pretraining Samples for Developing a Deep Learning Brain Injury Model via Transfer Learning", *Annals of Biomedical Engineering (ABME)*, 52(10):2726-2740. doi: 10.1007/s10439-023-03354-3.
- (12) Hong Zhang, Ming Liu, Jiashun Jin, **Wu, Zheyang** (2023) "Signal-noise ratio of genetic associations and statistical power of SNP-set tests", *Annals of Applied Statistics*. 17(3): 2410-2431 (September 2023). DOI: 10.1214/22-AOAS1725.
- (13) Liang Wang, Akiko Maehara, Xiaoguo Zhang, Rui Lv, Yangyang Qu, Xiaoya Guo, Jian Zhu, **Zheyang Wu**, Kristen L. Billiar, Jie Zheng, Lijuan Chen, Genshan Ma, Gary S. Mintz, Dalin Tang (2023) "Quantification of Patient-Specific Coronary Material Properties and Their Correlations with Plaque Morphological Characteristics: an In Vivo IVUS Study", *International Journal of Cardiology*, <https://doi.org/10.1016/j.ijcard.2022.09.051>.
- (14) Shaoju Wu, Wei Zhao, **Zheyang Wu**, Thomas McAllister, Jingwen Hu, Songbai Ji (2023) "Approximating Subject-specific Brain Injury Models via Scaling Based on Head-brain

- Morphological Relationships”, *Biomechanics and Modeling in Mechanobiology*, 22(1):159–175, PMID: 36201071 <https://doi.org/10.1007/s10237-022-01638-6>
- (15) Xiaohui Chen, Hong Zhang, Ming Liu, Hong-Wen Deng, **Wu, Zheyang** (2022) “Simultaneous detection of novel genes and SNPs by adaptive p-value combination”, *Frontiers in Genetics*. <https://doi.org/10.3389/fgene.2022.1009428>.
 - (16) Hong Zhang, **Zheyang Wu** (2022) “The General Goodness-of-fit Tests for Correlated Data”, *Computational Statistics & Data Analysis*, Vol. 167. 107379. <https://doi.org/10.1016/j.csda.2021.107379>.
 - (17) Hong Zhang, **Zheyang Wu** (2022) “The Generalized Fisher’s Combination and Accurate p-Value Calculation under Dependence”, *Biometrics*. <https://doi.org/10.1111/biom.13634>.
 - (18) Hong Zhang, Judong Shen, **Zheyang Wu** (2021) “An efficient and accurate approximation to the distribution of quadratic forms of Gaussian variables”, *Journal of Computational & Graphical Statistics*. <https://doi.org/10.1080/10618600.2021.2000423>.
 - (19) Janel O. Johnson, et. al., (my co-authorship through Familial ALS consortium) (2021) “Association of Variants in the SPTLC1 Gene With Juvenile Amyotrophic Lateral Sclerosis”, *JAMA Neurology*, doi:10.1001/jamaneurol.2021.2598. <https://jamanetwork.com/journals/jamaneurology/article-abstract/2783665>.
 - (20) Wei Zhao, **Zheyang Wu**, Songbai Ji (2021) “Displacement error propagation from embedded markers to brain strain”, *Journal of Biomechanical Engineering*. 143(10):101001. doi: 10.1115/1.4051050
 - (21) Han Yu, Pedro J. del Nido, Tal Geva, Chun Yan, **Zheyang Wu**, Rahul H. Rathod, Xueying Huang, Kristen L. Billiar, Dalin Tang (2021) “A Novel Pulmonary Valve Replacement Surgery Strategy Using Contracting Band for Patients with Repaired Tetralogy of Fallot: An MRI-Based Multi-Patient Modeling Study”, *Frontiers in Bioengineering and Biotechnology*. <https://doi.org/10.3389/fbioe.2021.638934>.
 - (22) Liang Wang, Dalin Tang, Akiko Maehara, **Zheyang Wu**, Chun Yang, David Mucigrosso, Mitsuaki Matsumura, Jie Zheng, Richard Bach, Kristen L. Billiar, Gregg W Stone, and Gary S. Mintz (2020) “Using intravascular ultrasound image-based fluid-structure interaction models and machine learning methods to predict human coronary plaque vulnerability change”, *Computer Methods in Biomechanics and Biomedical Engineering*, <http://doi.org/10.1080/10255842.2020.1795838>.
 - (23) Han Yu, Pedro J del Nido, Tal Geva, Chun Yang, **Zheyang Wu**, Rahul H Rathod, Xueying Huang, Kristen L Billiar, Dalin Tang (2020) “Multi-Band Surgery for Repaired Tetralogy of Fallot Patients with Reduced Right Ventricle Ejection Fraction: A Pilot Study”, *Frontiers in Physiology*, published: 19 March 2020 doi: 10.3389/fphys.2020.00198.
 - (24) Hong Zhang, Jiashun Jin, **Zheyang Wu** (2020) “Distributions and Power of Optimal Signal-Detection Statistics in Finite Case”, *IEEE Transactions on Signal Processing*. Vol. 68, 1021–1033. <https://ieeexplore.ieee.org/document/8961113>.
 - (25) Hong Zhang, Tiejun Tong, John E. Landers, **Zheyang Wu** (2020) “TFisher: A powerful truncation and weighting procedure for combining p-values”, *Annals of Applied Statistics*. 14(1): 178–201. <http://dx.doi.org/10.1214/19-AOAS1302>

- (26) Lei Ding, Hannah Schluter, Matthew Szcus, Rushdy Ahmad, **Zheyang Wu**, Weifeng Xu (2020) "Comparison of statistical tests and power analysis for phosphoproteomic data," *Journal of Proteome Research*. doi: 10.1021/acs.jproteome.9b00280.
- (27) Karen L. Troy, Megan E. Mancuso, Joshua E. Johnson, **Zheyang Wu**, Thomas J. Schnitzer, Tiffany A. Butler (2020) "Bone Adaptation in Adult Women Is Related to Loading Dose: A 12-Month Randomized Controlled Trial", *Journal of Bone and Mineral Research*. 10 March 2020 <https://doi.org/10.1002/jbmr.3999>
- (28) Stephen J. Foley, **Zheyang Wu**, Samuel M. Politz (2019) "A C. elegans MAP kinase pathway is required for wild-type display of an L1-specific surface antigen (srf-6 is nsy-1 III)," *microPublication Biology – Research Results*. 10.17912/micropub.biology.000129. Published: 2019/07/04.
- (29) Han Yu, Dalin Tang, Rahul H. Rathod, Tal Geva, Chun Yang, **Zheyang Wu**, Xueying Huang, Kristen L. Billiar, Pedro J. del Nido (2019) "Ventricle Stress/Strain Comparisons between Tetralogy of Fallot Patients and Healthy Using Models with Different Zero-Load Diastole and Systole Morphologies," *PLoS One*. 14;14(8):e0220328. doi: 10.1371/journal.pone.0220328. eCollection 2019.
- (30) Qingyu Wang, Dalin Tang, Liang Wang, Gador Canton, **Zheyang Wu**, Thomas S. Hatsukami, Kristen L. Billiar, Chun Yuan (2019), "Combining Morphological and Biomechanical Factors for Optimal Carotid Plaque Progression Prediction: An MRI-Based Follow-Up Study Using 3D Thin-Layer Models", *International Journal of Cardiology*. 15; 293: 266-271. doi: 10.1016/j.ijcard.2019.07.005.
- (31) Liang Wang, Dalin Tang, Akiko Maehara, David Molony, Jie Zheng, Habib Samady, **Zheyang Wu**, Wenbin Lu, Jian Zhu, Genshan Ma, Don P. Giddens, Gary S. Mintz (2019) "Multi-Factor Decision-Making Strategy for Better Coronary Plaque Burden Increase Prediction: A Patient-Specific 3D FSI Study Using IVUS Follow-Up Data", *Biomechanics and Modeling in Mechanobiology*. 2019, Oct;18(5):1269-1280. 2019 Apr 1. doi: 10.1007/s10237-019-01143-3
- (32) Qingyu Wang, Dalin Tang, Gador Canton, Thomas S. Hatsukami, Kristen L. Billiar, **Zheyang Wu**, Chun Yan (2019), "Impact Of Patient-Specific In Vivo Vessel Material Properties On Carotid Atherosclerotic Plaque Stress/Strain Calculations", *International Journal of Computational Methods*. 16(3). <https://doi.org/10.1142/S0219876218420021>.
- (33) Qingyu Wang, Dalin Tang, Gador Canton, **Zheyang Wu**, Thomas S. Hatsukami, Kristen L. Billiar, Chun Yuan (2018) "Stress-based Plaque Vulnerability Index and Assessment for Carotid Atherosclerotic Plaques Using Patient-Specific Vessel Material Properties", *Molecular & Cellular Biomechanics*. 15(4):189-201.
- (34) Han Yu, Pedro J. del Nido, Tal Geva, Chun Yang, Alexander Tang, **Zheyang Wu**, Rahul Rathod, Xueying Huang, Kristen Billiar, Dalin Tang (2019) "Patient-Specific In Vivo Right Ventricle Material Parameter Estimation for Patients with Tetralogy of Fallot Using MRI-Based Models with Different Zero- Load Diastole and Systole Morphologies", *International Journal of Cardiology*. 276: 93–99.
- (35) Yunliang Cai, Shaoju Wu, Wei Zhao, Zhigang Li, **Zheyang Wu**, Songbai Ji (2018) "Concussion classification via deep learning using whole-brain white matter fiber strains",

PLoS ONE. **13**(5): e0197992. <https://doi.org/10.1371/journal.pone.0197992>

- (36) Dalin Tang, Heng Zuo, Chun Yang, **Zheyang Wu**, Xueying Huang, Rahul H. Rathod, Alexander Tang, Kristen L. Billiar, Tal Geva (2017) "Comparison of Right Ventricle Morphological and Mechanical Characteristics for Healthy and Patients with Tetralogy of Fallot: An In Vivo MRI-Based Modeling Study". *Molecular & Cellular Biomechanics*. **14**(3): 137–151.
- (37) Liang Wang, Akiko Maehara, **Zheyang Wu**, Chun Yang, David Muccigrosso, Jie Zheng, Richard Bach, Kristen L. Billiar, Gary S. Mintz, Dalin Tang (2018), "Fluid-Structure Interaction Models Based on Patient-Specific IVUS at Baseline and Follow-Up for Prediction of Coronary Plaque Progression by Morphological and Biomechanical Factors: A Preliminary Study", *Journal of Biomechanics*. **68**: 43–50. doi: 10.1016/j.jbiomech.2017.12.007.
- (38) Qingyu Wang, Gador Canton, Jian Guo, Xiaoya Guo, Thomas S. Hatsukami, Kristen L. Billiar, Chun Yuan, **Zheyang Wu**, Dalin Tang (2017), "MRI-based patient-specific human carotid atherosclerotic vessel material property variations in patients, vessel location and long-term follow up", *PLoS ONE*. **12**(7): e0180829. <https://doi.org/10.1371/journal.pone.0180829>
- (39) Allen H. Hoffman, Zhongzhao Teng, Jie Zheng, **Zheyang Wu**, Pamela K. Woodard, Kristen L. Billiar, Dalin Tang (2017), "Stiffness Properties of Adventitia, Media and Full Thickness Human Atherosclerotic Carotid Arteries in the Axial and Circumferential Directions", *Journal of Biomechanical Engineering*. **139**(12). DOI: 10.1115/1.4037794.
- (40) Longling Fan, Jing Yao, Chun Yang, **Zheyang Wu**, Di Xu, and Dalin Tang (2016) "Material Stiffness Parameters as Potential Predictors of Presence of Left Ventricle Myocardial Infarction: 3D Echo-Based Computational Modeling Study". *BioMedical Engineering Online*, 2016 Apr 5;15:34. doi: 10.1186/s12938-016-0151-8.
- (41) Dalin Tang, Chun Yang, Pedro J. del Nido, Heng Zuo, Rahul H. Rathod, Xueying Huang, Vasu Gooty, Alexander Tang, Kristen L. Billiar, **Zheyang Wu**, and Tal Geva (2016) "Mechanical stress is associated with right ventricular response to pulmonary valve replacement in patients with repaired tetralogy of fallot". *The Journal of Thoracic and Cardiovascular Surgery*. **151**(3):687-694.e3. doi: 10.1016/j.jtcvs.2015.09.106.
- (42) Dalin Tang; Pedro J del Nido; Chun Yang; Heng Zuo; Xueying Huang; Rahul R Rathod; Vasu Gooty; Alexander Tang; **Zheyang Wu**; Kristen L Billiar; Tal Geva (2016) "Patient-Specific MRI-Based Right Ventricle Models Using Different Zero-Load Diastole and Systole Geometries for Better Cardiac Stress and Strain Calculations and Pulmonary Valve Replacement Surgical Outcome Predictions". *PLoS ONE* 11(9):e0162986.
- (43) Dongni Zhang, Hongzhu Cui, Dmitry Korkin, and **Zheyang Wu** (2015) "Incorporation of protein binding effects into likelihood ratio test for exome sequencing data". *BMC Proceedings*. 9 Suppl 8:S37.
- (44) Liang Wang, Jie Zheng, Akiko Maehara, Chun Yang, Kristen L. Billiar, **Zheyang Wu**, Richard Bach, David Muccigrosso, Gary S. Mintz, and Dalin Tang (2015) "Morphological and Stress Vulnerability Indices for Human Coronary Plaques and Their Correlations with Cap Thickness and Lipid Percent: An IVUS-Based Fluid-Structure Interac-

- tion Multi-Patient Study". *PLoS Computational Biology*. **11**(12):e1004652. eCollection 2015. doi: 10.1371/journal.pcbi.1004652.
- (45) Elizabeth T. Cirulli, et. al. (2015) "Exome sequencing in amyotrophic lateral sclerosis identifies risk genes and pathways", *Science*. **347**(6229): 1436-1441. DOI: 10.1126/science.aaa3650. (Coauthor ship as part of FALS Sequencing Consortium).
 - (46) Liang Wang, **Zheyang Wu**, Chun Yang, Jie Zheng, Richard Bach, David Muccigrosso, Kristen Billiar, Akiko Maehara, Gary S. Mintz , and Dalin Tang (2015), "IVUS-Based FSI Models for Human Coronary Plaque Progression Study: Components, Correlation and Predictive Analysis", *Annals of Biomedical Engineering*. **43**(1): 107-121. DOI: 10.1007/s10439-014-1118-1.
 - (47) Bradley N. Smith, Nicola Ticozzi, Claudia Fallini, Athina Soragia Gkazi, Simon Topp, Kevin P. Kenna, Emma L. Scotter, Jason Kost, Pamela Keagle, Jack W. Miller, Daniela Calini, Caroline Vance, Eric W. Danielson, Claire Troakes, Cinzia Tiloca, Safa Al-Sarraj, Elizabeth A. Lewis, Andrew King, Claudia Colombrita, Viviana Pensato, Barbara Castellotti, Jacqueline de Belleruche, Frank Baas, Anneloor LMA ten Asbroek, Peter C. Sapp, Diane McKenna-Yasek, Russell L. McLaughlin, Meraida Polak, Seneshaw Asress, Jesus Esteban-Perez, Jose Luis Mun oz-Blanco, Michael Simpson, SLAGEN Consortium, Wouter van Rheenen, Frank P. Diekstra, Giuseppe Lauria, Stefano Duga, Stefania Corti, Cristina Cereda, Lucia Corrado, Gianni Soraru, Karen E. Morrison, Kelly L. Williams, Garth A. Nicholson, Ian P. Blair, Patrick A. Dion, Claire S. Leblond, Guy A. Rouleau, Orla Hardiman, Jan H. Veldink, Leonard H. van den Berg, Ammar Al-Chalabi, Hardev Pall, Pamela J. Shaw, Martin R. Turner, Kevin Talbot, Franco Taroni, Alberto Garcia-Redondo, **Zheyang Wu**, Jonathan D. Glass, Cinzia Gellera, Antonia Ratti, Robert H. Brown, Jr., Vincenzo Silani, Christopher E. Shaw, and John E. Landers (2014), "Exome-wide rare variant analysis identifies TUBA4A mutations associated with familial ALS", *Neuron*. **84**(2):324-31. doi: 10.1016/j.neuron.2014.09.027.
 - (48) **Zheyang Wu**, Yiming Sun, Shiquan He, Judy Cho, Hongyu Zhao, and Jiashun Jin (2014), "Detection Boundary and Higher Criticism for Weak and Sparse Genetic Effects", *The Annals of Applied Statistics*. **8**(2): 824-851. DOI: 10.1214/14-AOAS724.
 - (49) Marisa Mariani, Mark McHugh, Marco Petrillo, Steven Sieber, Shiquan He, Mirko Andreoli, **Zheyang Wu**, Paul Fiedler, Giovanni Scambia, Shohreh Shahabi, and Cristiano Ferlini (2014), "HGF/c-Met Axis Drives Cancer Aggressiveness in the Neo-adjuvant Setting of Ovarian Cancer", *Oncotarget*. **5**(13):4855-67.
 - (50) Marisa Mariani, Shiquan He, Mark McHugh, Mirko Andreoli, Deep Pandya, Steven Sieber, **Zheyang Wu**, Paul Fiedler, Shohreh Shahabi and Cristiano Ferlini (2014), "Integrated Multidimensional Analysis is Required for Accurate Prognostic Biomarkers in Colorectal Cancer", *PLoS One*. **9**(7):e101065. doi: 10.1371/journal.pone.0101065.
 - (51) Jing Xuan, Li Yang, **Zheyang Wu** (2014), "Higher Criticism Approach to Detect Rare Variants Using Whole Genome Sequencing Data", *BMC Proceedings*. **8**(Suppl 1):S14. doi:10.1186/1753-6561-8-S1-S14.
 - (52) Yiwei Liu, Jing Xuan, **Zheyang Wu** (2014), "Extended T^2 Tests for Longitudinal Family Data in Whole Genome Sequencing Studies", *BMC Proceedings*. **8**(Suppl 1):S40.

doi:10.1186/1753-6561-8-S1-S40.

- (53) Li Yang, Jing Xuan, **Zheyang Wu** (2014), "A Goodness-of-Fit Association Test for Whole Genome Sequencing Data", *BMC Proceedings*. **8**(Suppl 1):S51. doi:10.1186/1753-6561-8-S1-S51.
- (54) Taoye Chen, Patchara Santawisook, **Zheyang Wu** (2014), "A Multi-level Model for Analyzing Whole Genome Sequencing Family Data with Longitudinal Traits", *BMC Proceedings*. **8**(Suppl 1): S86. doi:10.1186/1753-6561-8-S1-S86
- (55) **Zheyang Wu**, Yijuan Hu, Phillip E. Melton (2014), "Longitudinal Data Analysis for Genetic Studies in the Whole-Genome Sequencing Era", *Genetic Epidemiology*. **38**(S1), S74 – S80.
- (56) Dalin Tang, Chun Yang, Gador Canton, **Zheyang Wu**, Thomas Hatsukami and Chun Yuan (2013), "Correlations between Carotid Plaque Progression and Mechanical Stresses Change Sign over Time: A Patient Follow Up Study Using MRI and 3D FSI Models", *BioMedical Engineering*. **12**(105). doi: 10.1186/1475-925X-12-105.
- (57) Jing Zhang, **Zheyang Wu**, Chao Gao, and Michael Q. Zhang (2012), "High-Order Interactions in Rheumatoid Arthritis Detected by Bayesian Method Using GWAS Data", *The American Medical Journal*. **3**(1): 56-66.
- (58) Sakthikumar Ambady, **Zheyang Wu** and Tanja Dominko (2012) "Identification of Novel MicroRNAs in *Xenopus Laevis* Metaphase II Arrested Eggs", *Genesis*. **50**: 286-299.
- (59) **Zheyang Wu** and Huibin Zhou (2013), "Model Selection and Sharp Asymptotic Minimality", *Probability Theory and Related Fields*. **156**(1-2): 165-191. doi: 10.1007/s00440-012-0424-5
- (60) **Zheyang Wu** and Hongyu Zhao (2012), "On Model Selection Strategies to Identify Genes Underlying Binary Traits Using Genome-Wide Association Data". *Statistica Sinica*. **22**(3): 1041-1074.
- (61) Alun Thomas, Haley J Abel, Yanming Di, Laura L Faye, Jing Jin, Jin Liu, **Zheyang Wu** and Andrew Paterson (2011), "Effect of Linkage Disequilibrium on the Identification of Functional Variants". *Genetic Epidemiology*. **35**: S115–S119.
- (62) **Zheyang Wu**, Chun Yang and Dalin Tang (2011), "In Vivo Serial MRI-Based Models and Statistical Methods to Quantify Sensitivity and Specificity of Mechanical Predictors for Carotid Plaque Rupture: Location and Beyond". *Journal of Biomechanical Engineering*. **133**(6):064503-1.
- (63) Shiquan He and **Zheyang Wu** (2011), "Gene-Based Higher Criticism Methods for Large Scale Exonic SNP Data". *BMC Proceedings*. **5**(Suppl 9):S65.
- (64) **Zheyang Wu**, Hongyu Zhao (2009), "Statistical Power of Model Selection Strategies for Genome-Wide Association Studies", *PLoS Genetics* **5**(7): e1000582. doi:10.1371/journal.pgen.1000582.
- (65) **Zheyang Wu**, Chatchawit Apornthewan, David Ballard, Ji Young Lee, Joon Sang Lee, and Hongyu Zhao (2009), "Two-Stage Joint Selection Method to Identify Candidate Markers from Genome-Wide Association Studies," *BMC Proceedings*. **3**(Suppl 7):S29.

- (66) David Ballard, Chatchawit Aporn Dewan, Ji Young Lee, Joon Sang Lee, **Zheyang Wu**, and Hongyu Zhao (2009), "A Pathway Analysis Applied to GAW 16 Genome Wide Rheumatoid Arthritis Data," *BMC Proceedings*. 3(Suppl 7):S91.
- (67) Chatchawit Aporn Dewan, David Ballard, Ji Young Lee, Joon Sang Lee, **Zheyang Wu**, and Hongyu Zhao (2009), "Gene Hunting of the Genetic Analysis Workshop 16 Rheumatoid Arthritis Data Using Rough Set Theory", *BMC Proceedings*, 3(Suppl 7):S126.
- (68) Jeremiah D. Schuur, Akash Shah, **Zheyang Wu**, Howard Forman, and Cary P. Gross (2009), "The Impact of Medicaid Coverage and Reimbursement on Access to Diagnostic Mammography", *Cancer*. 115(23): 5566 - 5578.
- (69) Zhide Fang, Douglas P. Wiens, and **Zheyang Wu** (2006), "Locally D-optimal Designs for Multistage Models and Heteroscedastic Polynomial Regression Models", *Journal of Statistical Planning and Inference*. 136(11):4059-4070.
- (70) Brian L. Weiss, Rosa Mouchotte, Rita V. M. Rio, Yi-neng Wu, **Zheyang Wu**, Abdelaziz Heddi, and Serap Aksoy (2006), "Interspecific Transfer of Bacterial Endosymbionts between Tsetse Fly Species: Infection Establishment and Effect on Host Fitness", *Applied and Environmental Microbiology*. 72(11):7013-7021.

Book chapters:

- (71) Song Huang, David Ballard, **Zheyang Wu**, Hongyu Zhao (2007), "Expression quantitative trait loci mapping", *Current Topics in Human Genetics, Studies in Human Diseases*, edited by Hong-Wen Deng, Hui Shen, Yong-Jun Liu, Hai Hu. World Scientific Publishing Co Pte Ltd. Printed in Singapore. Page 379-431.

Dissertation thesis:

- (72) **Zheyang Wu**, "Model Selection Methods for High-dimensional Data and Their Applications to Genome-Wide Association Studies", Yale University, New Haven, CT

Refereed conference papers: 28 refereed conference papers and abstracts (2011–2019), primarily from cardiovascular and brain biomechanics collaborations; full list available on request.

4. Software packages

- *DPComb*: R package for Discrete p-Value Combination Tests. Available on CRAN: <https://CRAN.R-project.org/package=DPComb>.
- *Qapprox*: R package for approximation to the survival functions of quadratic forms of Gaussian variables. Available on CRAN <https://cran.r-project.org/web/packages/Qapprox/index.html>
- *GFisher*: R package for a generalized Fisher's p-value combination method. Available on CRAN <https://CRAN.R-project.org/package=GFisher>
- *TFisher*: R package for thresholding Fisher's p-value combination method. Available on CRAN: <https://cran.r-project.org/web/packages/TFisher/index.html>.

- *SetTest*: R package for group testing procedures for signal detection and goodness-of-fit tests. Available on CRAN: <https://cran.r-project.org/web/packages/SetTest/index.html>.
- *MarkerSearchPower*: R package for calculating statistical power of detecting disease markers based on model selection strategies.

5. Fellowships and grants received

Grants received:

- (1) PI, WPI Summer Sandbox Grant (teaching), "Structured AI Tutoring for Applied Statistics", \$5,800, Summer 2026. <https://www.wpi.edu/academics/undergraduate/summer-courses/summer-sandbox-awardees>
- (2) PI, NSF Statistics Program, NSF-2515791, "Innovations in Integrative Association Testing for Large Genetic Data", \$174,999.00, 08/15/2025 - 07/31/2028.
- (3) Co-Investigator, NIH/DHHS, 4UH3NS135168, "IMPACT: Integrative Mindfulness-Based Predictive Approach for Chronic Low Back Pain Treatment (UH3)", PI: Jean A King. \$6,889,914.00, 09/21/2023 - 08/31/2028.
- (4) Co-Investigator, NIH/DHHS, 1R15AR085912, "Foot biomechanics and metatarsal structure in runners with minimal and cushioned shoes", PI, Karen Troy. \$546,159.00, 08/01/2025 - 07/31/2028.
- (5) Consultant, Canadian Institutes of Health Research (CIHR), Project Grant, "Sex-specific biomechanical thresholds and cumulative risk models for sports-related brain injury", PI: Lyndia (Chun) Wu, University of British Columbia. \$1,470,000 CAD, newly awarded (dates being finalized).
- (6) Co-PI, The Longitude Prize on ALS Discovery Award, "The ALS Genome-AI Initiative", newly awarded, details forthcoming. <https://als.longitudeprize.org/discovery-award-teams>
- (7) Co-Investigator, NIH/DHHS, 5R21NS136884, "Biomechanical basis of sex differences in concussion and subconcussion". PI: Songbai Ji. \$391,804.00. 06/01/2024-05/31/2026.
- (8) PI, NSF Statistics Program, DMS-2113570, "New Techniques to Combine Measures of Statistical Significance from Heterogeneous Data Sources with Application to Analysis of Genomic Data", \$200,000, 08/01/2021 - 07/31/2025.
- (9) Co-Principal Investigator, NIH/DHHS, UG3NS135168, "IMPACT: Integrative Mindfulness-Based Predictive Approach for Chronic low back pain Treatment". Principal Investigator: Jean A King. \$1,643,229.00, 09/21/2023 - 09/20/2025.
- (10) Co-Investigator, NIH-NICHHD, 1R15HD104169-01A1, "Biomechanical factors affecting metatarsal fatigue and bone stress injury risk". PI: Karen Troy at WPI. \$557,351.00. 09/01/2021 - 08/31/2024.
- (11) PI, NSF Statistics Program, DMS-1812082, "Optimal and Adaptive p-Value Combination Methods with Application to ALS Exome Sequencing Study", \$150,000, 07/01/2018 - 06/30/2022.

- (12) PI, NSF Statistics Program, DMS-1309960, "Optimal tests for weak, sparse, and complex signals with application to genetic association studies", \$109,999, 08/15/2013 - 07/31/2017.
- (13) PI, Sponsored Research Agreement, Data and Statistical Sciences Department, AbbVie Inc. "Patient Subgrouping through Interactive Associations", \$48,354, 08/15/2017 - 05/05/2018.
- (14) PI, Subcontract of NIH grant from University of Massachusetts Medical School, "Analysis of deep sequencing data to identify genes causative for neurodegenerative diseases", \$93,174, 04/01/2012 - 03/31/2014.
- (15) Co-PI, NSF DMS-1337943, "MRI: Acquisition of High-Performance Computing System for Research, Education and Training", \$363,579, 09/01/2013 - 07/31/2017 (PI: Homer Walker at WPI).
- (16) Co-investigator, NIH NHLBI, 1R15HL121761-01A1, "Arrhythmia detection using a smart-phone", \$388,706, 08/01/2014 - 07/31/17. (PI: Ki H. Chon at WPI).
- (17) Co-investigator, NIH NCI, 5R01CA166379-02, "MR-Guided Precision Conformal Ablation Therapy for Brain Tumors", \$3,790,423, 09/01/2013 - 04/30/2018. (PI: Gregory Fischer at WPI).
- (18) Co-investigator, NIH, 5R01AR063691-03, "A Prospective Study of Human Bone Adaptation Using a Novel In-Vivo Loading Model", 08/16/2013 - 08/31/2017. \$1,436,413. (PI: Karen Troy at WPI).
- (19) Senior Personnel, National Science Foundation, DMS-1263127, "REU Site: Research Experiences for Undergraduates in Industrial Mathematics and Statistics", \$382,000 , 05/01/13 - 04/30/16 (PI: Suzanne L. Weekes).
- (20) Senior Personnel, National Science Foundation, DMS-9732338, "REU Site: Research Experiences for Undergraduates in Industrial Mathematics and Statistics", \$333,000, 05/01/10 - 04/30/13 (PI: Suzanne L. Weekes).

6. Presentations at conferences, seminars, and colloquia

- (1) Invited Talk: "Discreteness in Statistical Genetics: p-Value Combination, Feasible Dependence, and Genotype Simulation", Department of Mathematics, Hong Kong Baptist University, Hong Kong, June 30, 2026.
- (2) Invited Talk: "GTsim: Feasible Correlations of Discrete Random Variables and Genotype Data Simulation", International Chinese Statistical Association (ICSA) Annual Meeting, Shenzhen, China, June 28-30, 2026.
- (3) Invited Talk: "Optimal Weighting for Integrative Association Tests: Application to Whole-Genome Sequencing Studies", April 11, 2025, Biogen CoLab, Cambridge, MA.
- (4) Invited Talk: "An optimal weighting approach for enhancing integrative association tests in whole-genome sequencing studies (WGS)", 2024 Southeast University Statistics and Big Data Forum, Nanjing, China, July 15-17, 2024.

- (5) Invited Talk: “Enhancing Integrative Association Tests: Optimal Weighting Approaches in Whole-Genome Sequencing (WGS) Studies”, 2nd Joint Conference on Statistics and Data Science (JCSDS), Kunming, China, July 12-14, 2024.
- (6) Invited Lecture (6 hours): “Statistical Hypothesis Testing: From Fundamental Principles to Frontier Research”, Series Short Courses in JCSDS 2024, Yunnan University, Kunming, China, July 6-11, 2024.
- (7) Invited Talk: “Asymptotic Correlation Robustness of Supremum-based P-value Combination Tests”, 2024 New England Statistics Symposium (NESS), University of Connecticut, Storrs, CT, May 20 - 24, 2024.
- (8) Contributed Talk: “A minimum Wasserstein distance approach to Fisher’s combination of independent discrete p-values”, Joint Statistical Meetings (JSM), Toronto, Canada, August 5-10, 2023.
- (9) Poster: “Simultaneous detection of novel disease-associated DNA segments and SNPs by data-adaptive p-value combination”, Poster, 2023 Human Genetics and Genomics by Gordon Research Conference, Waterville Valley, New Hampshire, July 9-14, 2023.
- (10) Invited Talk: “Optimal Weighting for Integrative Tests of Genetic Associations”, 2023 NESS (New England Statistics Symposium), Boston University, MA, June 3 - 6, 2023.
- (11) (Invited) “Signal-noise ratio of genetic associations and statistical power of SNP-set tests”, 2022 ICSA Applied Statistic Symposium, Gainesville, Florida June 19-22, 2022.
- (12) (Invited) “P-value Combination Tests for Correlated Data Analysis”, Department of Statistics and Probability Colloquium, Michigan State University, October 29, 2019.
- (13) “P-value combination methods for integrative meta-analysis of genome-wide associations”, Poster, American Society of Human Genetics Annual Meeting, Houston, TX, October 15–19, 2019.
- (14) (Invited) “A P-value Combination Framework for Integrative Meta-Analysis of Genome-Wide Associations”, International Chinese Statistical Association (ICSA) China Conference, Nankai University, Tianjin, China. July 1–4, 2019.
- (15) (Invited) “P-value Combination Tests for Correlated Data Analysis”, Department of Mathematical Sciences Colloquium, Michigan Tech, April 5, 2019.
- (16) “Novel Optimal and Adaptive P-value Combination Tests”, Poster, American Society of Human Genetics Annual Meeting, San Diego, October 16–20, 2018.
- (17) (Invited) “Optimal and Data-Adaptive P-value Combination Tests For Correlated Data”, Tsinghua University, Beijing, China. July 30, 2018.
- (18) (Invited) “Novel Optimal and Data-Adaptive P-value Combination Tests”, 2018 International Chinese Statistical Association China Conference, Qingdao, China. July 2–5, 2018.
- (19) (Invited) “Optimal and Data-Adaptive P-value Combination Tests For Correlated Data”, Yunnan University of Finance and Economics, July 9, 2018.
- (20) (Invited) “Optimal and Data-Adaptive P-value Combination Tests For Correlated Data”, The Hong Kong University of Science and Technology, Department of Electronic & Computer Engineering, June 23, 2018.

- (21) (Invited) "Optimal and Data-Adaptive P-value Combination Tests For Correlated Data", Hong Kong Baptist University, Department of Mathematics, June 22, 2018.
- (22) (Invited) "Optimal and Adaptive P-value Combination Methods", The 2nd International Conference on Econometrics and Statistics (EcoSta 2018), City University of Hong Kong, June 19– 21, 2018.
- (23) (Invited) "On Statistical Association and Prediction Methods for Study of Coronary Plaque Progression", Southeastern University, Nanjing, China, June 19, 2018.
- (24) (Invited) "The gGOF and tFisher: Optimal and Data-Adaptive P-value Combination Tests With Applications in Omics Data Analysis", The 1st International Symposium on Genomics and Translational Medicine, Suzhou, P. R. China, Jun 14–17, 2018.
- (25) "Optimal Association Tests for Finding Weak Genetic Effects", Poster at IMA Workshop of Innovative Statistics and Machine Learning for Precision Medicine, September 14 – 16, 2017, University of Minnesota.
- (26) (Invited) "Signal Detection and Goodness-Of-Fit Tests For Correlated Data", 61st ISI World Statistics Congress (WSC), July 16 – 21, 2017, Marrakech, Morocco.
- (27) (Invited) "Statistical Detection and Prediction when Signals are Weak, Rare, Correlated, and Interactive", April 25, 2017, AbbVie Inc., Worcester.
- (28) "Optimal Association Tests for Finding Weak Genetic Effects", Poster at Gordon Research Conferences Quantitative Genetics & Genomics – Exploiting Context and Increasing Predictive Value in Complex Trait Genetics, Feb 26 - Mar 3, 2017, Galveston, TX, United States
- (29) (Invited) "Optimal Methods for Detecting Weak and Sparse Signals Based on Correlated Features in Genetic Association Studies", 10th ICSA (International Chinese Statistical Association) International Conference, Shanghai Jiao Tong University, Shanghai, China, December 19-22, 2016.
- (30) (Invited) "Methods for Detecting Weak and Sparse Signals Among Correlated Features", Department of Mathematics, Northeastern University, November, 8, 2016.
- (31) "Detection Boundary and Optimal Association Tests", Poster at Human Genetics & Genomics, Gordon Research Conference, July, 2015.
- (32) (Invited) "Approximating the Distributions of Optimal Goodness-of-Fit Tests with Applications in GWAS", The 29th New England Statistical Symposium, University of Connecticut, April 24-25, 2015.
- (33) (Invited) "Patterns of genetic effects and the power of genetic association studies", Department of Electronic & Computer Engineering, HKUST, Hong Kong, January. 2015.
- (34) (Invited) "Optimal tests for weak and sparse signal detection and the calculation of their distributions", Departments of Mathematics, HKBU, Hong Kong, January. 2015.
- (35) (Invited) "Patterns of genetic effects and the power of genetic association studies", Departments of Mathematical Sciences, IUPUI, Sept. 2014.
- (36) "Detection Boundary and Optimal Association Tests for Rare and Weak Genetic Effects", Poster at Human Single Nucleotide Polymorphisms & Disease, Gordon Research Conference, August, 2014

- (37) (Invited) "Genetic Effects and Optimal Statistical Association Tests Using GWAS and Sequence Data", Departments of Bioinformatics and Computational Biology and Biostatistics, Brown University, April 2014.
- (38) (Invited) "Statistical Signal Detection For Hunting Genes Using High-Dimensional Genetic Data", Departments of Mathematics, Northeastern University, January 2014.
- (39) "Detection Boundary and Higher Criticism Approach for Rare and Weak Genetic Effects", American Society of Human Genetics Annual Meeting, Boston, MA, October 2013.
- (40) (Invited) "Genetic Effects and Statistical Power of Gene Hunting Using GWAS and Sequence Data", Department of Quantitative Health Sciences, Umass Medical School, September 2013.
- (41) "Truncated and weighted transformation methods for combining p-values", Joint Statistical Meetings, Montreal, Quebec, Canada, August 2013.
- (42) (Invited) "Truncated and weighted Z-transformation methods for combining p-values", New England Statistics Symposium, University of Connecticut, CT, April 2013.
- (43) "A multi-level model for analyzing whole genome sequencing family data with longitudinal traits", Genetic Analysis Workshop 18, Stevenson, WA, October, 2012.
- (44) (Invited) "Detection Boundary and Higher Criticism Approach for Sparse and Weak Genetic Effects", International Chinese Statistical Association Symposium, Boston, MA, June 2012.
- (45) (Invited) "Higher criticism for high dimensional regression with application to gene detection", International Conference and Exhibition on Biometrics & Biostatistics, Omaha, NE, March 2012.
- (46) "Detection Boundary and Higher Criticism Approach for Sparse and Weak Genetic Effects", New Researchers in Statistics and Probability, UCSD, CA, July 2012.
- (47) "Higher Criticism for Detecting Sparse and Weak Genetic Effects", Joint Statistical Meetings, San Diego, CA, July 2012.
- (48) "On model selection strategies to identify genes underlying binary traits", Joint Statistical Meetings, Miami, FL, August 2011.
- (49) (Invited) "Higher Criticism for High Dimensional Data with Application to Genome-wide Association Studies", New England Statistics Symposium, University of Connecticut, CT, April 2011.
- (50) (Invited) "Penalized model selection and asymptotic minimaxity", International Conference on Statistics and Society, Beijing, China, July 2010.
- (51) (Invited) "Model selection methods for high-dimensional data and their applications to genome-wide association studies", University of Connecticut Department of Statistics, March 2010.
- (52) "Power of genome-wide search strategies for binary trait loci", American Society of Human Genetics Annual Meeting, DC, November 2010.

- (53) "Power of genome-wide search strategies for binary trait loci", International Genetic Epidemiology Society Meeting, Boston, MA, October, 2010.
- (54) "Gene-based Higher Criticism methods for large scale exonic SNP data", Genetic Analysis Workshop 17, Boston, MA, October, 2010.
- (55) "A genome-wide association testing method incorporating linkage disequilibrium", Joint Statistical Meetings, Vancouver, British Columbia, Canada, August, 2010.
- (56) "Model selection and sharp asymptotic minimaxity", Post Presentation, Innovation and Inventiveness in Statistics Methodologies, Yale University, New Haven, CT, May 2009.
- (57) (Invited) "Model selection methods for high-dimensional data and their applications to genome-wide association studies", Department of Statistics, West Virginia University, WV. March 2009.
- (58) (Invited) "Model selection methods for high-dimensional data and their applications to genome-wide association studies", Division of Biostatistics at New York University School of Medicine, NY. March 2009.
- (59) (Invited) "Model selection methods for high-dimensional data and their applications to genome-wide association studies", Department of Preventive Medicine, Northwestern University, IL. March 2009.
- (60) (Invited) "Model selection methods for high-dimensional data and their applications to genome-wide association studies", Department of Mathematical Sciences, Worcester Polytechnic Institute, MA. April 2009.
- (61) "Gene detection and regression", Department of Mathematical Sciences, Worcester Polytechnic Institute, MA. April 2009.
- (62) (Platform Presentation) "Power of three model selection methods for genome-wide association data", American Society of Human Genetics Annual Meeting, Philadelphia, PA, November 2008.
- (63) "Power of model selection methods for genome-wide association studies," Joint Statistical Meetings, Denver, CO, August 2008.
- (64) "Locally D-optimal designs for multistage models and heteroscedastic polynomial regression models," Louisiana Chapter of the ASA, New Orleans, LA, June 2004.

7. Patents awarded or pending

- (1) Co-inventor, U.S. Provisional Application Serial No.: 61/489,075 "Computational Modeling for a Atherosclerotic Plaque Rupture" (Inventor: Dalin Tang at WPI).

8. Consulting

- (1) Grave Instruments Incorporated (Statistical Consulting 2011).

SERVICE

9. Grant proposal review for funding agencies

- (1) NSF Panelist for DMS/Statistics Program, 2026 March.
- (2) Reviewer for The Israel Science Foundation, on the submissions of Personal Research Grants. 2025 May.
- (3) NSF Panelist for DMS/Emerging Mathematics in Biology (eMB), 2023.
- (4) NSF Panelist for DMS/MRI, 2021
- (5) NSF Panelist for DMS/NIGMS, 2020
- (6) National Security Agency, Mathematical Sciences Program, Grants for Research in the Mathematical Sciences, 2016
- (7) Kentucky Science and Engineering Foundation, Research & Development Excellence Program, 2016

10. WPI administrative leadership, committee assignments, department or campus-wide

Campus level service:

- (1) Member of Steering Committee, Bioinformatics and Computational Biology (BCB) Program, August 2009 - Present.

Service at Department of Mathematical Sciences:

- (1) Member, Departmental Tenure Committee (Elected), 2026 Spring.
- (2) Member, Departmental Undergraduate Committee, 2010, 2013, 2014, 2024, 2025 Fall.
- (3) Member, Departmental Teaching Evaluation Committee for Prof. Charlotte Fowler, 2025 Fall.
- (4) Co-chair, Graduate Program Committee, August 2023 – July 2024.
- (5) Member, Hiring Committee for Tenure track AP in Biostatistics/Statistics/Epidemiology, 2023-2024.
- (6) Member, Hiring Committee for Statistics Teaching Faculty, 2022-2023.
- (7) Member, Graduate Program Committee, August 2011 - August 2012, August 2019 – 2022.
- (8) NTT Statistics faculty searching committee: 2020-2021.
- (9) Served as a Professional Associate for faculty promotions: 2016, 2017, 2020.
- (10) Member, Graduate Admission Committee, August 2014 - May 2015, August 2017 - August 2019
- (11) Teaching evaluation committee for peers: 2016, 2020.
- (12) Professional Development Workshop Series – Applying for Jobs in Academia, Fall 2017
- (13) Chair for PhD Preliminary Exams: Liang Wang (Applied Mathematics); Lu Chen (Statistics); Hong Yan (Statistics)

- (14) Member, Center for Industrial Mathematics and Statistics, October 2009 - 2017.
- (15) Member, WPI Bioinformatics Faculty Hiring Committee, 2010-2011, 2012 - 2013.
- (16) Co-coordinator, Project Presentation Day, 2012.
- (17) Judge, WPI Graduate Research Achievement, 2012.
- (18) Member, Colloquium Committee, 2009 - 2010.

Service for Statistics program:

- (1) Organizer, designer and grader: GCE Exams in Probability and in Mathematical Statistics I and II (MA540, MA541), 2011 - Present.
- (2) Co-organizer for Statistical Seminar. 2016 –.
- (3) Admission of new students for the Master in Applied Statistics program and PhD in Statistics program.
- (4) Contributions to new hirings for faculty positions in statistics and related to statistics.
- (5) Supporting the annual statistical alumni meetings.

11. Memberships and offices held in professional societies

Professional society membership:

- (1) American Statistical Association (ASA)
- (2) Institute of Mathematical Statistics (IMS)
- (3) The American Society of Human Genetics (ASHG)
- (4) International Chinese Statistical Association (ICSA)

12. Conference organization, editorial and referee services

Journals served peer-reviews:

- (1) Genes
- (2) Entropy
- (3) Biometrics
- (4) Bioinformatics
- (5) Human Genetics and Genomics (HGG) Advances
- (6) Frontiers in Statistical Genetics and Methodology
- (7) BMC Bioinformatics
- (8) Statistica Sinica
- (9) IEEE/ACM Transactions on Computational Biology and Bioinformatics
- (10) Mathematical Reviews
- (11) Scientific Reports
- (12) Electronic Journal of Statistics

- (13) Annals of Statistics
- (14) Journal of the American Statistical Association
- (15) Annals of Applied Statistics
- (16) IEEE Transaction on Signal Processing
- (17) Journal of Statistical Planning and Inference
- (18) Statistical Analysis and Data Mining
- (19) Annals of the Institute of Statistical Mathematics
- (20) Statistics and Its Interface
- (21) Statistical Methodology
- (22) Journal of the Korean Statistical Society
- (23) Statistical Applications in Genetics and Molecular Biology
- (24) Annuals of Human Genetics
- (25) Genetics
- (26) Human Genetics
- (27) Genomics
- (28) BMC Research Notes
- (29) BMC Genetics
- (30) BMC Medical Genomics

Conference organization:

- (1) Organizer and Chair, International Chinese Statistical Association (ICSA) Annual Meeting, Shenzhen, China, June 28-30, 2026. Sessions:
 - Statistical Methods for Genetics and Genomics
 - Statistical Learning and Modeling for Complex Data
 - Advanced Methods for Dependence, Signals and Spatio-Temporal Detection
- (2) Scientific Committee Member, The 1st International Symposium on Genomics and Translational Medicine, Suzhou, China, June 14-17, 2018
- (3) Co-organizer of Phylogenetics and Statistical Genetics Section, The 29th New England Statistical Symposium, University of Connecticut, April 24-25, 2015.
- (4) Scientific and Organizing Committee Member, Post-WCB Workshop: Vulnerable Plaques: Data, Modeling, Predictions and Translation to Clinical Applications, Worcester, July 12-13, 2014.
- (5) Organizer and Chair, Statistical Genetics and Genomics Session, International Chinese Statistical Association (ICSA) annual meeting, Boston, 2012.
- (6) Reviewer, The Tenth Asia Pacific Bioinformatics Conference, Melbourne, Australia, 2012.

- (7) Program Committee Member, The 3rd International Conference on BioMedical Engineering and Informatics, Yantai, China, 2010.
- (8) Chair, Asymptotic Theory Session, International Conference on Statistics and Society, Beijing, China, 2010

Conference mentoring programs:

- (1) Served as a mentor in the Docent/Mentoring Program for first time attendees in the Joint Statistical Meetings (JSM), in 2016 and 2017.
 - (2) Served as a mentor in the Woman Researcher Mentoring program in Gordon Research Conferences: Quantitative Genetics & Genomics in 2017.
13. Significant civic, cultural, religious, and similar contributions
- (1) April 2019 – April 2020, Stroke and Turn Judge, New England Swimming / USA Swimming.

TEACHING

14. Teaching innovations at WPI

- (1) **Cofounder of Bioinformatics and Computational Biology (BCB) Program at WPI.** Co-created BS, BS/MS, MS, and PhD programs in BCB.
- (2) **Statistics program development:** Co-created the new PhD program in Statistics.
- (3) **New course creation:** BCB504/BCB4004/MA584/MA4603 Statistical Methods in Genetics and Bioinformatics. This is a foundational course for the BS, BS/MS, MS, and PhD degrees in Bioinformatics and Computational Biology (BCB). It is also cross-listed as a Mathematics (MA) course for undergraduate and graduate students, which opens a potential new direction in biostatistics.
- (4) **Course redesign and sharing:** MA2611 Applied Statistics I. New teaching plan has been created based on a new design of 36 topic modules and educational technology. Innovated teaching materials include module-based high quality lecture videos, note-taking-friendly slides in LaTeX, online homework assignments and exercises based on WeBWorK and Poll Everywhere, computer labs based on R Markdown, database of over 1,000 exercise problems and answers in LaTeX and WeBWorK formats. The new design well supports both on-campus and remote teaching. Materials available on Canvas for sharing. Similar teaching innovation work has been done for MA2610 Statistics in Life Sciences in 2020 B term.
- (5) **Course redesign and sharing:** MA2631 Probability. Innovated teaching has been created and implemented based on a new design of 54 topic modules. New teaching materials include 55 high-quality lecture videos and notes, 84 pages of lecture content in LaTeX, a database of 102 problems and answers in LaTeX for homework, sample exams, and exams. The new design well supports both on-campus and remote teaching. Materials have been shared with faculty on Canvas.

- (6) **Course redesign and sharing:** MA 542 - Regression Analysis: Developed 45 topic-based lecture modules, documented in LaTeX format, with recorded videos for each module shared on Canvas. Compiled 14 R example code files in RMarkdown format, 11 homework sets with questions and solutions in LaTeX format, and R solution in RMarkdown. All materials have been shared with instructors teaching this course.
- (7) **Course redesign and sharing:** BCB 4004/MA 4603/BCB 504/MA 584, Statistical Methods in Genetics and Bioinformatics: Created topic modules covering four areas: basic biology, basic probability and statistics, quantitative genetics, and statistics for bioinformatics. This includes 30 lectures recorded in videos, accompanied by lecture notes in LaTeX format and 14 coding example files in RMarkdown format. Compiled 6 homework sets, 2 quizzes, and one written exams, with questions and solutions in LaTeX and R solution in RMarkdown. Based on these new teaching materials, separate undergraduate and graduate courses are planned to be offered starting next academic year. All materials have been shared on Canvas.
- (8) **Course redesign for BCB4004/MA4603 Statistical Methods in Genetics and Bioinformatics; 2024.** Previously, this course shared the title and course numbers BCB504 / BCB4004 / MA584 / MA4603, cross-listed for undergraduate and graduate students and taught simultaneously. However, due to significant growth in enrollment (35 students in the 2024 A term), we have decided to separate the undergraduate and graduate courses starting in 2024. The newly designed undergraduate course covers three main topics: basic molecular genetics, statistical and probabilistic foundations, and basic human population genetics. Each topic includes updated lecture notes, examples, homework questions, quizzes, and exams. The current undergraduate course features 29 topic modules and has been significantly enhanced with coding examples in RMarkdown format.
- (9) **Course Redesign and Sharing: BCB504/MA584 Statistical Methods in Genetics and Bioinformatics (2025).** Redesigned a formerly cross-listed undergraduate/graduate course into a rigorously structured, graduate-only offering aligned with advanced training needs in statistics and quantitative genetics. The revision elevated theoretical depth, expanded applied examples and coding components, and implemented a redesigned assessment framework (homework, quizzes, and a research-oriented class project). The teaching plan was restructured to fit a two-lectures-per-week, semester-long format. AI-assisted learning and research components were introduced and integrated into coursework and projects. Developed a full suite of new instructional materials, including LaTeX lecture notes, R Markdown coding examples, homework sets, quizzes, and project guidelines, all deployed on Canvas and ready for dissemination.

15. Courses taught at WPI (Lecture courses by term, title, and enrollment)

- (1) 2025 A Term, MA 2611-AL01 - Applied Statistics I, 2 sections, 50 students
- (2) 2025 Fall Semester, BCB 504-F01/ MA 584-F01 - Statistical Methods In Genetics And Bioinformatics, 9 students
- (3) 2025 Summer, MA 2611-E2-L01 - Applied Statistics I, 13 students

- (4) 2025 D Term, MA 2611-DL04 - Applied Statistics I, 2 sections, 62 students
- (5) 2025 Spring Semester, MA 542-S01 - Regression Analysis, 14 students
- (6) 2024 Fall A Term, MA 2611-AL01 - Applied Statistics I, 46 students
- (7) 2024 Fall A Term, BCB 4004-A01/ MA 4603-A01 - Statistical Methods In Genetics And Bioinformatics, 35 students
- (8) 2024 Spring D Term, MA 2611-DL03 - Applied Statistics I (group 3), 2 sections, 51 students
- (9) 2024 Spring D Term, MA 2611-DL02 - Applied Statistics I (group 2), 4 sections, 64 students
- (10) 2023 Fall A Term, BCB 4004-A01/ MA 4603-A01 - Statistical Methods In Genetics And Bioinformatics, 20 students
- (11) 2023 Fall A Term, MA 2611-AL02 (group 2) - Applied Statistics I, 3 sections, 66 students
- (12) 2023 Fall Semester, BCB 504-F01/ MA 584-F01 - Statistical Methods In Genetics And Bioinformatics, 13 students
- (13) 2023 Summer, MA 2611-E2-L01 - Applied Statistics I, 12 students
- (14) 2023 Spring D Term, MA 2611-DL02 - Applied Statistics I, 4 sections, 103 students
- (15) 2023 Spring, MA 542-S01 - Regression Analysis, 10 students
- (16) 2022-E MA2631 *Probability* (9 students)
- (17) 2022-D MA2611 *Applied Statistics I* (6 sections, 142 students)
- (18) 2022-C MA2611 *Applied Statistics I* (5 sections, 81 students)
- (19) 2021-Fall BCB540/BCB4004/MA584/MA4603 *Statistical Methods in Genetics and Bioinformatics* (4 sections, 29 students)
- (20) 2021-E MA2611 *Applied Statistics I* (25 students)
- (21) 2021-Spring MA541/MA4632 *Probability and Mathematical Statistics II* (2 sections, 12 students)
- (22) 2021-C MA2611 *Applied Statistics I* (4 sections, 83 students)
- (23) 2020-B MA2610 *Applied Statistics for Life Sciences* (2 sections, 48 students)
- (24) 2020-A MA2631 *Probability* (Combined section, 35 students)
- (25) 2020-E MA2611 *Applied Statistics I* (Combined section, 54 students)
- (26) 2019-Fall BCB540/BCB4004/MA584/MA4603 *Statistical Methods in Genetics and Bioinformatics* (4 sections, 16 students)
- (27) 2019-B MA2610 *Applied Statistics for Life Sciences* (3 sections, 59 students)
- (28) 2019-B MA2611 *Applied Statistics I* (6 sections, 165 students)
- (29) 2019-A MA2611 *Applied Statistics I* (5 sections, 131 students)
- (30) 2019-E MA2611 *Applied Statistics I* (2 sections, 35 students)
- (31) 2018-Fall MA546 *Design & Analysis of Experiments* (17 students)

- (32) 2018-B MA2610 *Applied Statistics for Life Sciences* (3 sections, 52 students)
- (33) 2018-B MA2611 *Applied Statistics I* (6 sections, 139 students)
- (34) 2018-A MA2611 *Applied Statistics I* (5 sections, 111 students)
- (35) 2018-E MA2611 *Applied Statistics I* (2 sections, 35 students)
- (36) 2018-Spring BCB510 *BCB Seminar* (5 registered students, 30+ audience on average)
- (37) 2018-Spring MA541/MA4632 *Probability and Mathematical Statistics II* (2 sections, 8 students)
- (38) 2018-D MA2611 *Applied Statistics I* (2 sections, 52 students)
- (39) 2017-Fall BCB540/BCB4004/MA584/MA4603 *Statistical Methods in Genetics and Bioinformatics* (4 sections, 28 students)
- (40) 2017-B MA2610 *Applied Statistics for Life Sciences* (3 sections, 49 students)
- (41) 2017-E MA2611 *Applied Statistics I* (2 sections, 24 students)
- (42) 2015-Fall BCB510 *BCB Seminar* (1 registered students, 30+ audience on average)
- (43) 2015-B MA2610 *Applied Statistics for Life Sciences* (3 sections, 37 students)
- (44) 2015-A MA2611 *Applied Statistics I* (4 sections, 99 students)
- (45) 2015-Fall BCB540/BCB4004/MA584/MA4603 *Statistical Methods in Genetics and Bioinformatics* (4 sections, 19 students)
- (46) 2015-E MA2611 *Applied Statistics I* (2 sections, 28 students)
- (47) 2014-Fall MA540/MA4631 *Probability and Mathematical Statistics I* (2 sections, 20 students)
- (48) 2014-E MA2611 *Applied Statistics I* (2 sections, 16 students)
- (49) 2014-Spring BCB540/BCB4004/MA584/MA4603 *Statistical Methods in Genetics and Bioinformatics* (4 sections, 29 students)
- (50) 2014-D MA2610 *Applied Statistics for Life Sciences* (3 sections, 68 students)
- (51) 2014-C MA2611 *Applied Statistics I* (4 sections, 60 students)
- (52) 2013-Fall MA540/MA4631 *Probability and Mathematical Statistics I* (2 sections, 24 students)
- (53) 2013-E MA2611 *Applied Statistics I* (2 sections, 10 students)
- (54) 2013-Spring MA541/MA4632 *Probability and Mathematical Statistics II* (2 sections, 26 students)
- (55) 2013-D MA2610 *Applied Statistics for Life Sciences* (2 sections, 36 students)
- (56) 2013-C MA2611 *Applied Statistics I* (4 sections, 91 students)
- (57) 2012-Fall MA546 *Design & Analysis of Experiments* (15 students)
- (58) 2012-D MA2610 *Applied Statistics for Life Sciences* (2 sections, 40 students)
- (59) 2011-Spring BCB540/BCB4004/MA584/MA4603 *Statistical Methods in Genetics and Bioinformatics* (7 students)

- (60) 2011-Fall MA540/MA4631 *Probability and Mathematical Statistics I* (24 students)
 - (61) 2011-Fall MA540/MA4631 *Probability and Mathematical Statistics I* (25 students)
 - (62) 2011-Spring MA546 *Design & Analysis of Experiments* (14 students)
 - (63) 2011-D MA2610 *Applied Statistics for Life Sciences* (2 sections, 57 students)
 - (64) 2011-C MA2611 *Applied Statistics I* (4 sections, 96 students)
 - (65) 2010-D MA2610 *Applied Statistics for Life Sciences* (2 sections, 43 students)
 - (66) 2010-C MA2611 *Applied Statistics I* (3 sections, 86 students)
 - (67) 2009-Fall MA546 *Design & Analysis of Experiments* (4 students)
16. Projects, theses and dissertations advised at WPI (summary; full lists available on request)
- PhD dissertations advised: 5
 - Master's theses advised: 7
 - Master's capstone / research projects advised: 19
 - Undergraduate Major Qualifying Projects (MQPs) advised: 8; Interactive Qualifying Projects (IQPs): 1
 - NSF-funded Research Experience for Undergraduates (REU) projects: 2
 - K-12 research mentorship projects (high school students and teachers): 6
17. Independent studies conducted at WPI: 73 independent-study and project courses supervised for graduate and undergraduate students (list available on request).
18. Academic advising and mentoring at WPI (summary)
- Mentor for junior faculty (2) and postdoctoral fellows (4)
 - Advisor for PhD students (6); PhD dissertation committee member (9)
 - Academic advisor for Master's students (34) and undergraduate students (31)
 - Coaching student fellowship applications (NSF GRFP)
19. Student awards won under my research mentorship
- Student travel award from The 19th Genetic Analysis Workshop, \$1,200, Aug. 24 - 27, 2014. Support my research team of one WPI student.
 - Student travel award from The 18th Genetic Analysis Workshop, \$3,600, Oct. 14 - 17, 2012. Support my research team of four WPI students.